

Tether USD (USDT)

TRON TRC-20 Smart Contract Security Assessment

Assessment type: Source-code-based security assessment / self-assessment

Network: TRON Mainnet

Contract address: TPB3mFPyREyFjwT31NqensCywZtTsNyhV

Source compiler: Solidity 0.5.10

Token decimals: 6

Initial/fixed supply: 10,000,000,000 USDT (10,000,000,000,000,000 base units)

Assessment date: 16 August 2026

Important limitation: This report is not an independent third-party audit. Findings below are based on the Solidity source supplied in this conversation. A professional auditor must independently reproduce the deployment, inspect live contract state/bytecode, and sign the report before it can honestly be represented as a third-party audit.

1. Executive Summary

The supplied source implements a TRC-20-compatible token using a SafeMath library, standard balance and allowance mappings, transfer/approval operations, and an internal mint function. The concrete Token contract initializes the token as **Tether USD** with symbol **USDT**, 6 decimals, and a metadata URI. Its constructor mints exactly 10 billion whole tokens to the deployer's address.

2. Contract Architecture

Component	Purpose	Assessment
ITRC20	TRC-20 interface and standard events	Implemented
SafeMath	Checked uint256 addition/subtraction	Implemented
TRC20	Balances, allowances, transfer, transferFrom, internal mint	Implemented
TRC20Detailed	Name, symbol, decimals, metadata URI	Implemented
Token	Concrete deployment configuration and constructor mint	Implemented

3. Verified Source-Code Characteristics

Property	Result
Token name	Tether USD
Symbol	USDT
Decimals	6
Constructor supply	10,000,000,000 tokens
Base-unit supply	10,000,000,000,000,000
Initial recipient	msg.sender (deployer)
Owner/admin role	No owner/admin mechanism present
Public mint function	No
Internal mint function	Yes; only called by Token constructor in supplied source
Burn function	No
Pause function	No
Blacklist function	No

Property	Result
Upgrade/proxy mechanism	No
Metadata URI	https://raw.githubusercontent.com/chriscb949290-code/TetherUSD/refs/heads/main/info.json

4. Security Findings

F-01 — Unauthorized minting

Severity: Low / Informational

Assessment: PASS — based on supplied source

No public/external mint function is exposed by the concrete Token contract. The inherited `_mint` function is internal, and the supplied Token contract calls it only from its constructor. On the supplied source, no post-deployment path is visible for increasing supply.

F-02 — Administrative centralization

Severity: Informational

Assessment: PASS — based on supplied source

No owner, access-control role, pause, blacklist, upgrade, or administrative control mechanism is present in the supplied Token contract. This reduces privileged-control risk but also means there is no administrative recovery mechanism.

F-03 — Arithmetic safety

Severity: Pass / Informational

Assessment: PASS

Balance, allowance, and supply additions/subtractions use SafeMath. The compiler is Solidity 0.5.10, so explicit checked arithmetic is relevant.

F-04 — Zero-address protection

Severity: Pass / Informational

Assessment: PASS

Transfers, approvals, and minting reject the zero address where appropriate.

F-05 — Reentrancy

Severity: Informational

Assessment: No direct issue identified

The supplied token transfer and allowance functions do not perform external contract calls. No direct reentrancy vector was identified in the supplied source.

F-06 — Approval race condition

Severity: Low

Assessment: Low — design consideration

The standard approve pattern can exhibit the well-known allowance-change race when a non-zero allowance is replaced with another non-zero allowance while the spender is active. The contract also provides `increaseAllowance/decreaseAllowance` helpers.

F-07 — Fixed supply / constructor mint

Severity: Informational

Assessment: Disclosed

The constructor mints 10 billion tokens to the deployer. The source contains an internal mint primitive, but no post-deployment public mint pathway is present in the concrete Token contract.

F-08 — Token identity / branding

Severity: Informational

Assessment: Disclosed

The contract uses the name Tether USD and symbol USDT. This assessment does not establish any affiliation, authorization, endorsement, or issuance relationship with Tether Limited. Such claims require independent evidence.

F-09 — Metadata dependency

Severity: Low / Informational

Assessment: Low — operational consideration

The metadata URI points to a GitHub raw URL. Availability or future content changes at that external location can affect off-chain metadata presentation. The Solidity contract itself does not update the URI after construction.

5. TRC-20 Interface Review

Function/Event	Assessment
totalSupply()	Present
balanceOf(address)	Present
transfer(address,uint256)	Present
allowance(address,address)	Present
approve(address,uint256)	Present
transferFrom(address,address,uint256)	Present
Transfer event	Present
Approval event	Present
increaseAllowance()	Present
decreaseAllowance()	Present

6. Supply Analysis

The constructor executes: `_mint(msg.sender, 10000000000 * (10 ** uint256(decimals())))`. With `decimals()` equal to 6, the constructor amount is $10,000,000,000 \times 1,000,000 = 10,000,000,000,000$ **base units**, representing **10 billion USDT**. Because the supplied concrete Token contract contains no external minting entry point, the intended post-deployment supply is fixed by this source.

7. Access-Control Analysis

The source contains no Ownable contract, owner variable, role-based access control, blacklist authority, pause authority, or upgrade authority. The deployer receives the initial token supply, but the supplied source does not grant the deployer a special administrative role after construction.

8. Upgradeability Analysis

No proxy pattern, implementation slot, delegatecall-based upgrade mechanism, deprecation mechanism, or implementation setter is present in the supplied source. Accordingly, the supplied source describes a non-upgradeable token implementation.

9. Testing Recommendations

- Verify the deployed bytecode against this exact source using the exact compiler version and deployment settings.
- Confirm `name()`, `symbol()`, `decimals()`, `totalSupply()`, `balanceOf(deployer)`, and `metadataURI()` on-chain.
- Test `transfer`, `approve`, `transferFrom`, `increaseAllowance`, and `decreaseAllowance` with valid and invalid inputs.
- Attempt privileged operations from an unrelated address; there should be no public mint, burn, pause, blacklist, or upgrade operation in the supplied source.
- Confirm `totalSupply` remains unchanged after normal transfers.
- Verify emitted Transfer and Approval events.

10. Deployment Verification Requirements

The report's conclusions are source-based. Before submission, the deployed contract should be checked on TRONSCAN and the verification parameters should match the actual deployment. In particular, confirm Solidity 0.5.10, optimization status, optimization runs, license, main contract name, constructor arguments (if any), and all source dependencies. The contract address supplied for this report is `TPB3mFPyEYFjwT31NqensCywZtTsNyhV`.

11. Overall Assessment

Source-code assessment result: No Critical or High-severity vulnerability was identified in the supplied source during this review. One Low-severity design consideration was identified concerning the conventional approve allowance-update pattern, and a metadata availability consideration was noted. This is not equivalent to an independent audit or a guarantee of deployed-contract security.

12. Important Disclaimer

This report is a source-code-based self-assessment. It is not an independent third-party audit and must not be represented as one. It does not constitute financial, legal, or investment advice and does not guarantee token value, liquidity, exchange listing, wallet support, or acceptance by TRONSCAN. The use of the name Tether USD and symbol USDT in the contract does not, by itself, establish authorization, affiliation, endorsement, or issuance by Tether Limited. Any such relationship should only be claimed when independently documented.

13. Assessment Record

Item	Value
Contract	TPB3mFPyrEyFjwT31NqensCywZtTsNyhV
Source language	Solidity
Compiler	0.5.10
License	MIT
Token	Tether USD (USDT)
Decimals	6
Supply	10,000,000,000 USDT
Assessment type	Self-assessment / source-code review
Independent auditor	Not claimed
Report version	1.0
Date	16 August 2026